



INSTITUTE FOR
STRATEGIC TECHNOLOGIES

REPORT

MICROELECTRONICS

Securing the Chip Supply: A National Microelectronics Strategy

Building resilient domestic semiconductor capacity for the next decade of strategic competition.

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EXECUTIVE SUMMARY

The strategic stakes of domestic chip capacity

Advanced semiconductors underpin every priority defense capability, from autonomous systems and secure communications to the sensors that enable precision effects. Yet the capacity to design and fabricate the most advanced nodes remains concentrated in a handful of firms and geographies, creating a structural vulnerability that adversary action or natural disruption could exploit with strategic consequence.

This report assesses the current state of the domestic microelectronics base and offers a roadmap for a durable national strategy. It argues that fabrication incentives alone are insufficient: a resilient ecosystem requires coordinated investment across design, advanced packaging, materials, and workforce, sustained across administrations and budget cycles.

"Resilience is not built in a single fab. It is built across the full stack — design, materials, packaging, and the people who master them."

A widening gap

Commercial demand for leading-edge capacity increasingly sets the terms on which defense programs can access advanced nodes. Without deliberate intervention, defense requirements risk being crowded out by higher-volume commercial buyers, lengthening timelines and raising costs for the systems that most depend on cutting-edge silicon.

ILLUSTRATIVE: DOMESTIC SHARE OF ADVANCED-NODE CAPACITY (INDEXED)



FINDINGS

Toward a full-stack strategy

The Institute's analysis points to four mutually reinforcing lines of effort. Each is necessary; none is sufficient on its own. Sequenced correctly, they compound — early investment in packaging and workforce raises the return on later fabrication capacity.

1. Advanced packaging as a first-order priority

As gains from transistor scaling slow, advanced packaging has become a primary lever for performance. A national strategy that treats packaging as an afterthought will forfeit much of the value of upstream fabrication investment.

2. Assured access for defense programs

Mechanisms that guarantee defense access to advanced capacity — without distorting commercial markets — are essential to translate industrial capacity into fielded capability on relevant timelines.

3. Materials and supply resilience

Substrates, specialty chemicals, and rare inputs represent chokepoints as consequential as fabrication itself. Diversifying and, where feasible, reshoring these inputs reduces the leverage available to potential adversaries.

4. Workforce as strategic infrastructure

The binding constraint on new capacity is increasingly talent. A sustained pipeline — spanning technicians, engineers, and researchers — is the difference between capacity on paper and capacity in production.